

FLIES



THEY MAY BE PESTS AT TIMES, but flies are remarkable insects. As their name suggests, they have mastered the power of flight. Fast and agile in the air, flies dart about, hover, and make lightning turns. There are about 90,000 different insects that we call flies. About 75,000 of these are true flies, which have only one pair of wings and belong to the insect group Diptera. The remainder form many other groups of insects with two pairs of wings. Unlike most other types of fly, the larvae of true flies are completely different from the adults. Often called maggots, they have simple bodies with no legs and are little more than eating machines.

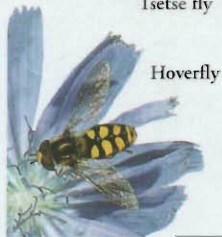
Feeding

Fly larvae feed on foods such as microscopic organisms, living flesh, plants, and dung. The mouthparts of adult flies are adapted for a liquid diet. They have extendible tubes to draw fluids into their bodies. The feeding habits of flies cause many health problems worldwide, from stomach upsets to more serious illnesses such as cholera.

Abdomen swollen with blood



Tsetse fly



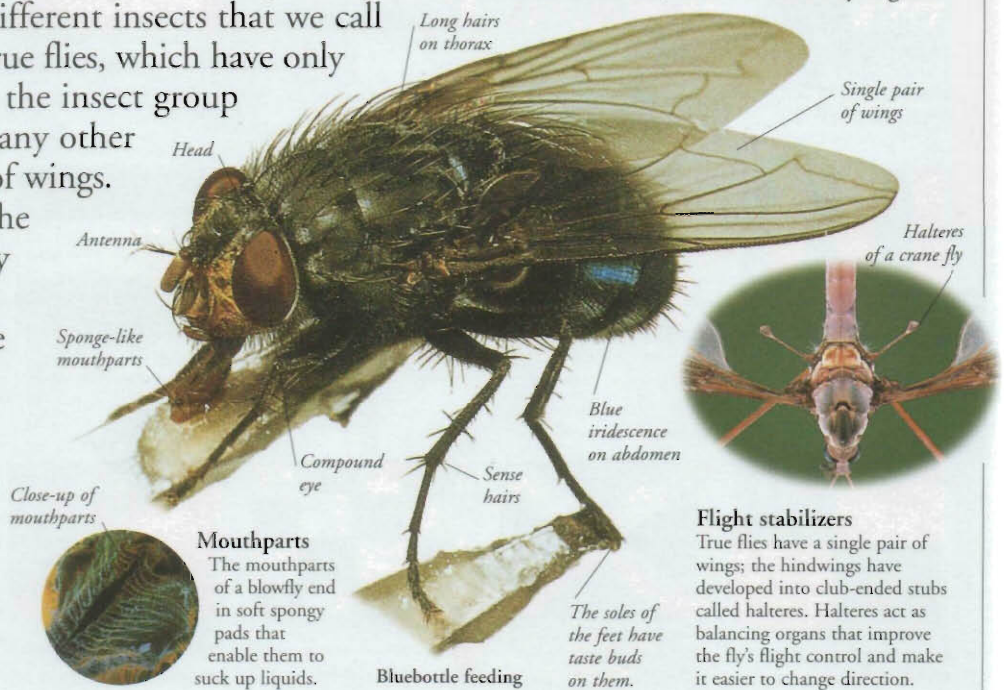
Hoverfly

Bloodsuckers and predators

Bloodsuckers and flies that catch prey have piercing mouthparts that cut holes in their victims. They inject anti-clotting agents to keep blood flowing, or poison to kill the prey. Enzymes are also released to help break down the body contents.

Nectar and waste feeders

Flies that feed on nectar or decaying matter have soft pads on the ends of their sucking mouthparts, that help soak up liquid food. On solid food, flies deposit saliva, then suck up the partly digested juice that results.



Breeding

After mating, female flies lay hundreds of eggs on a suitable feeding site for the larvae, that develop from the eggs. These sites may be in dung, soil, or water, or on leaves, dead bodies, or living animals. The larvae eat voraciously, and grow into pupae within which they change into their adult shape and form.



Robber flies mating



Bluebottle larvae

Mating

Before mating, flies may go through elaborate courtship rituals. Fruit flies dance on leaves, and gnats dance in the air. After mating, female robber flies commonly eat their mates.

Larvae

Fly eggs often hatch out into larvae on dead animals, which they begin to eat. The larvae, also called maggots, may live longer than the adult flies into which they develop.

Aquatic larvae

Mosquitos and many other bloodsucking flies lay their eggs in water. The larvae that develop float upside-down on the surface. They breathe through a tube attached to the abdomen, that pokes above the water. After pupation, the adult emerges to fly away.



Mosquito larva

Types of fly

In addition to true flies, many other kinds of fly exist that all have two pairs of wings, such as dragonflies and mayflies. The young that emerge from their eggs have a more complicated body structure than the larvae of true flies – some even look like wingless versions of the adults.

Dragonflies

Dragonflies are large predators. They dart around in search of other insects that they catch in flight with their long legs. They lay their eggs in water.



Giant compound eyes

Caddis flies

Adult caddis flies always live near water. Their larvae live underwater, and carry with them a protective case made of plant debris or sand.



Long antenna

Mayflies

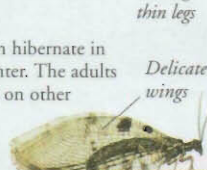
Young mayflies live below water. After growing wings, they leave the water and form swarms in the air. The adults mate, lay their eggs, then die a few hours later.



Long, thin legs

Lacewings

Lacewings often hibernate in houses over winter. The adults and larvae prey on other insects such as aphids. They are weak fliers.



Delicate wings

BLUEBOTTLE

SCIENTIFIC NAME	<i>Calliphora erythrocephala</i>
ORDER	Diptera
FAMILY	Calliphoridae
DISTRIBUTION	Europe
HABITAT	Fields, meadows, houses, and buildings
DIET	Rotting flesh, faeces, and other decomposing organic matter; adults also eat nectar
SIZE	Length 10 mm (0.4 in)
LIFESPAN	Larvae: 7 days; pupae: 8–10 days; adults: unknown

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